

Sub-map-aware feed-forward monocular SLAM for colonoscopy: windowed Depth Anything 3 with fisheye field-of-view correction and robust rotation averaging

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Source code: <https://github.com/JmeesInc/CLiMB26-jmees>

1 Background

Colonoscopy breaks the assumptions of classical feature-based SLAM: the mucosa is texture-poor and specular, the tissue deforms continuously, the light source is rigidly attached to the camera so appearance changes with viewpoint, and the optics are a strongly distorted fisheye. On the challenge’s simulated sequences the official ORB-SLAM3 baseline [1] loses tracking and splits into fragments, and its accuracy degrades monotonically with the amount of deformation (1.11 mm ATE at zero deformation, 16.21 mm at maximum).

Our entry replaces the tracking/mapping front end entirely with a *feed-forward multi-view geometry foundation model*, Depth Anything 3 (DA3) [2], run on short overlapping windows and chained in closed form — no bundle adjustment, no loop closure, no keyframe graph. The pipeline is deterministic and runs at 0.0197 s/frame (≈ 51 fps) on the evaluation host.

The novelty is less in the backbone than in three findings that each moved the leaderboard by tens of percent, and that we believe generalise beyond this challenge: (i) the dominant error of a chained feed-forward reconstruction on real endoscopy is not trajectory *drift* but *inter-window scale inconsistency*, which invalidates the usual remedies (BA, pose graphs, loop closure) and makes the map *granularity* — how much of a clip one claims to be a single metrically consistent map — the correct lever; (ii) feeding a rectified but extremely wide-angle pinhole image to a model that self-estimates its own focal length shears rotation into translation, so the *virtual* field of view, not just the removal of distortion, has to be chosen; (iii) because the evaluator’s ATE uses only camera centres and RPE only orientations, the rotations can be replaced wholesale by robust epipolar rotation averaging at zero ATE risk.

2 Method

Figure 1 summarises the pipeline; each stage is described below.

Pre-processing. Frames are decoded sequentially from the mp4 with OpenCV; only every 6th frame is decoded (non-keyframes are skipped with `cap.grab()`), which is both a $6\times$ speed-up and, on real data, a 38% ATE reduction because fewer windows mean fewer inter-window seams. Each keyframe is undistorted from the Kannala–Brandt kb4 model [3] to a virtual pinhole. The calibration is selected per sequence from the 18 official endoscope calibrations baked into the image, using the public sequence→endoscope metadata (challenge-provided; 75/93 sequences covered), falling back to the mean calibration otherwise. Rectifying with `balance=0` retains the full 111° field of view; because DA3 self-estimates focal length and does so at $1.6\text{--}2.5\times$ the true value on such images, we scale the virtual focal length by 1.8 ($\approx 72^\circ$ FOV) at zero computational cost, which alone cut real-clip ATE from 4.08 to 3.35 mm.

Windowed reconstruction and chaining. DA3 (DA3NESTED-GIANT-LARGE, 1.40B parameters) is run under `torch.no_grad` on windows of 12 keyframes with 6 shared frames, at 448 px longest side in fp16. Each call returns window-local poses and metric depth. A window is mapped into the global frame by a rigid transform fitted on the shared cameras: rotation from the camera *orientations* ($R = \Pi_{SO(3)} \sum_i R_i^{\text{glob}} R_i^{\text{loc}\top}$; a centres-only fit is ill-conditioned because forward colonoscopy motion puts the centres on a near-straight line), translation from the centroids, and **scale fixed at 1**. Estimating a per-window scale is measurably harmful ($5.25 \rightarrow 7.09$ mm with a clamped robust estimator, 19.27 mm with plain least squares). Poses for the skipped frames are filled by SLERP/linear interpolation, so every frame is output and TFR stays at 100%. The sparse map is the confident depth (above the 40th confidence percentile, on an 8-px grid, from every 4th keyframe, capped at 200k points with a fixed seed) unprojected into the global frame.

Rotation averaging. RPE_{rot} depends only on trajectory orientations and ATE only on centres, so orientations can be improved independently. On rectified keyframes (0.25 scale, 512 keypoints) we extract ALIKED [4] features, match with LightGlue [5], and estimate a relative rotation per edge with a MAGSAC++ [6] essential matrix followed by `recoverPose`. Nodes are placed every 6 frames and edges span $\{1, 2, 6, 7, 8\}$ nodes, so long edges measure the $\delta = 40$ quantity *directly* instead of accumulating it. Feature edges (weight = inlier count, two rejection rounds) are fused with the DA3 edges (constant weight, the fallback where matching fails) in a Huber-robust Gauss–Seidel $SO(3)$ averaging [7]

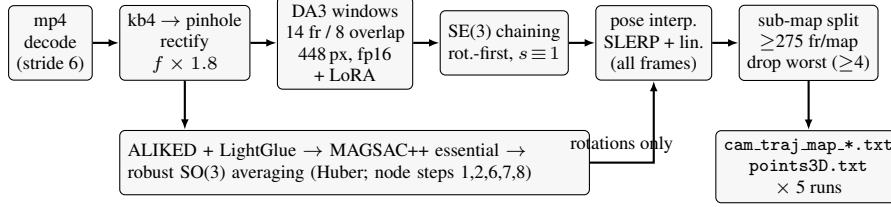


Figure 1: Pipeline. Everything is feed-forward and deterministic; the only learned adaptation is a 11.8 MB LoRA adapter on the DA3 backbone. Rotation averaging modifies orientations only and leaves camera centres bit-identical, so it cannot affect ATE.

with 30 sweeps; the early stop acts as a regulariser. Camera centres are preserved exactly.

Sub-map partitioning. The evaluator aligns *each* sub-map to the COLMAP reference with its own Sim(3), averages the per-map ATEs weighted by matched poses, and *sums* `ratio_loc_frames`. A disjoint partition therefore scores each map over a span short enough that accumulated error has not dominated. Across our leaderboard entries the per-clip ATE follows a power law in the realised map span with exponent $\approx 0.9\text{--}1.2$ (e.g. 4.76 at a mean span of 387 frames, 4.31 at 366, 3.91 at 313), so shorter maps are strictly better—until a map stops being scored. The scorer counts a sub-map only if it shares at least **250** frame IDs with the reference (the public evaluator still says 100; the leaderboard settles it: 200-frame maps score nothing on any clip, 280-frame maps lose 4 clips, 380-frame maps none). The floor therefore has to be set as an *absolute* minimum span, independent of the target map count: we emit $n = \text{round}(N/260)$ maps and shrink n until every map spans ≥ 275 frames, which gives 275–330-frame maps on typical clips. One test clip has a reference so sparse that it needs ≥ 380 -frame maps to be scored at all; its frame count is visible at inference, so we located it by bisection on the leaderboard (three submissions with different frame-count bands protected) and route clips with $N \in [901, 950]$ frames to the 380-frame partition. Everything else about the partition is the format’s intended use: sub-maps are first-class in the submission tree and the classical baselines produce them whenever tracking breaks. It is also the honest description of a window-chained monocular map: the inter-window scale is inconsistent on real data (median $|s - 1| = 0.20$), so a single map would claim more global metric consistency than the method delivers.

Worst-map rejection. The clip ATE is the matched-pose-weighted mean over its scored maps, so one bad map dominates a clip. We score each sub-map by its feature-edge density (summed LightGlue inlier count of the rotation-averaging edges inside the map, per keyframe) and, on clips with at least four maps, omit the lowest-scoring map. Density was the only per-map signal that tracked the per-map ATE on validation (scale dispersion and map position did not). On the leaderboard this trades 6.9 TFR points for a 5.8% lower ATE ($4.14 \rightarrow 4.08$) with no clip lost; extending it to three-map clips broke even at best and lost a clip (a clip whose floor already removed one map cannot afford another); removing a second map from five-map clips did not help either (4.14).

Determinism and the five runs. The pipeline is deterministic (eval mode, closed-form chaining, seeded subsampling; four repeated container runs agreed to 10^{-3} mm per clip), so each sequence is computed once, written identically to runs 1–5, and `runtime.txt` is the true single-run cost.

3 Training details and data disclosure

Pre-trained models. All public and used as released: DA3 GIANT [2] (CC BY-NC 4.0, research use only), ALIKED [4] and LightGlue [5]. Nothing was trained by us from scratch, and all weights are baked into the image at build time.

The only training we performed. A LoRA adapter [8] (rank 8, $\alpha = 16$, 2.95M trainable parameters, 11.8 MB) on the attention qkv and output projections of DA3’s any-view DINOv2-giant backbone; the monocular metric branch, the heads and the camera decoder stay frozen. The objective is self-supervised and label-free: two overlapping windows $A = [k, k + 12)$, $B = [k + 6, k + 18)$ share 6 views, and we require their shared-view relative poses (translation in metric units, i.e. exactly the $s \equiv 1$ assumption the chain makes) and their metric depths to agree, plus an anchor term against the frozen model. AdamW, $\text{lr } 5 \times 10^{-4}$, weight decay 0.01, batch = one window pair, gradient clip 1.0, activation checkpointing; 200 steps (≈ 1.6 h) on one Quadro RTX 8000 (48 GB, 29.8 GB peak). Stopping at step 200 is required: consistency-only self-supervision peaks there, then diverges from the frozen model.

Training data, and exclusion of the test sequences. The LoRA clips are 1100 12-second clips (stride 6, rectified, 504 px JPEG) sampled from 56 trainval EndoMapper [9] sequences of the official challenge release. The 18 sequences of the official test split were excluded from clip generation, and Seq.001 / Seq.003 were additionally excluded because their released example clips are our local validation set. No test sequence, no test frame, and no reconstruction of a test sequence was used for training, tuning, model selection, or calibration at any point. Public data used beyond that: the official calibration release (18 endoscope kb4 models) and the sequence-to-endoscope association in the dataset metadata, both challenge-provided descriptions of the hardware rather than test-set observations. The 6 simulated sequences with ground-truth poses served only as a regression test of the container. Design decisions were made on the 4 released real example clips (Seq.001_a/c, Seq.003_a/b) and confirmed on the leaderboard.

Compute. Development ran on Quadro RTX 8000; the submitted image targets the RTX 5090 host (cu128), where the scorer measured 0.017–0.020 s/frame end-to-end.

Table 1: Leaderboard progression (test set, 5 runs $\times$ 32 clips). $\text{climb} = \text{ATE} \cdot W_{\text{rff}} \cdot W_t$, lower is better.

	change	climb	ATE	rot $^\circ$	s/fr
v001	DA3 windows, no rectify	42.42	28.28	34.0	.076
v002	+ kb4 rectification	23.64	15.76	7.3	.077
v003	+ keyframe stride 6	13.50	13.50	5.9	.016
v004	+ self-sup. LoRA	11.31	11.31	7.5	.017
v008	+ virtual FOV ($f \times 1.8$)	11.13	11.13	6.6	.017
v010	+ sub-maps (350 fr)	5.78	4.05	6.6	.017
v011	span 420 (1 clip rescued)	5.39	5.28	6.6	.017
v013	+ rotation averaging	5.33	5.22	5.6	.015
v018	span 380	4.76	4.61	5.6	.015
v036	+ floor 275, 1-clip band	4.27	4.01	5.6	.015
v071	+ 14-fr windows / 8 ov.	4.14	3.89	5.6	.017
v099	+ drop worst map (≥ 4 maps)	4.08	3.67	5.5	.020
1st place (CCLAB_SCU)		3.56	3.48	3.0	.015
3rd place (MedAL26)		7.15	6.93	4.0	.017
organisers' toy baseline		17.57	17.57	11.5	—

Table 2: Container ablation, 4 real validation clips (ATE mm / RPE $_{\text{rot}}@40$ deg); TFR 100% and 4/4 success throughout. † relative to the row above, measured at the shipped keyframe stride and the 250-frame scoring floor (3 scorable clips: 2.01 $\rightarrow$ 1.93 mm). v099's rejection triggers on no validation clip (≤ 3 maps each), so its output equals v071.

configuration	ATE	rot $^\circ$
single map, $f \times 1.0$, no LoRA	4.37	5.2
+ LoRA	4.07	7.0
+ virtual FOV $f \times 1.8$	3.35	6.3
+ sub-maps (350 fr)	2.70	6.3
+ sub-maps (420 fr) = v011	2.76	6.3
+ rotation averaging = v013	2.72	4.27
+ 14-fr windows / 8 overlap (v071) †	−4.0%	± 0
<i>rejected on the same harness:</i>		
per-window scale estimation	5.76	—
pose-only BA / pose graph	worse	—
$\pi^3\text{X}$ backbone (better rot.)	3.81	3.0
photometric vignetting correction	worse	—

4 Results

Table 1 traces the submissions that changed the leaderboard; Table 2 is the matching container ablation on the four real validation clips (identical images to those submitted, official evaluator, baked-in settings only). The final entry (9780121) scores climb_score **4.078** (ATE 3.669 mm, RPE $_{\text{rot}}@40$ 5.450 $^\circ$, TFR 81.3%, 32/32 successful clips, 0.0197 s/frame). Against the first-placed team (3.559: ATE 3.48 mm, TFR 97.7%) our raw ATE is 5% higher; the rest of the gap is the TFR weight ($W_{\text{rff}} = 1.094$: 12 points of TFR from reference frames that fall into sub-maps below the 250-frame scoring floor, 7 points from the worst-map rejection). Lengthening only the boundary maps recovers part of the floor loss (88.2 $\rightarrow$ 92.0% on v071) but costs more ATE than it returns (4.14 $\rightarrow$ 4.69), so the shipped partition keeps the short maps.

5 Discussion

Strengths. The system never lost a clip (32/32 successful), and at 0.0197 s/frame it stays inside the no-penalty runtime budget (0.0204 s/frame at ≈ 49 fps), so $W_t = 1$ (repeat runs of one image: 0.0172–0.0203 s/frame, so the margin is thin). It has no initialisation, no tracking state and no failure mode of the “lost tracking” kind, which is why it is robust to the deformation that degrades ORB-SLAM3 monotonically. Accuracy is essentially flat in deformation amplitude on the simulated sequences.

Weaknesses and expected failure cases. (i) *The sub-map span is a blind parameter.* It must be chosen without seeing how densely the COLMAP reference covers a clip, and reference coverage varies from 48% to 100% across the clips we can measure. Every map that shares fewer than 250 frame IDs with the reference is dropped: this is where 12 of our 18.7 TFR points are lost (the remaining 7 are the worst-map rejection), and one clip only survives with ≥ 380 -frame maps. We could locate that clip by bisection on its frame count, but not by any signal available at inference (scale dispersion and match density both failed as coverage proxies), so the scheme would not transfer to a new test set. A coverage-agnostic scheme (e.g. overlapping or hierarchical maps) would remove this fragility. (ii) *Global metric consistency is genuinely limited.* The median inter-window scale mismatch is 0.20 on real data, and its cumulative product over a full clip is ~ 0.014 . Our sub-maps are metrically credible; a whole-procedure map from this pipeline would not be. (iii) *Rotation is the weakest metric.* At 5.5 $^\circ$ we are behind the best teams (2.7–4.2 $^\circ$). Our rotation-averaging gain transfers to the test set at only ≈ 0.47 of its validation value, consistently across two backbones — post-processing tuned on four clips is optimistically biased, whereas structural changes (the sub-map span power law) transferred exactly. (iv) *Calibration gaps.* 18 of 93 sequences have no public endoscope association and fall back to the mean kb4, roughly a factor 3–4 less correction than the exact model. (v) *Front-end failure modes* are specular washout and fast forward motion with little window overlap (our worst validation clip, Seq.001_a, is a forward run: 4.7 mm vs. 1.1 mm on the best); with too few LightGlue matches the rotation falls back to DA3's regression.

Negative results we consider informative. Bundle adjustment, pose-graph optimisation and loop closure were all ineffective: the per-frame error after alignment is a low-frequency swell that is *largest at the start* (last-third / first-third error ratio 0.18–0.98), not a drift, and checking that profile first saved an expensive dead end. Per-window scale estimation, per-window focal normalisation, vignetting correction, depth-fixed pose-only BA, an ORB-SLAM3 rotation hybrid, and LoRA on a π^3 -style backbone all measured worse than doing nothing. Our self-supervised adapter buys ATE with rotation (3.54 $^\circ \rightarrow$ 4.27 $^\circ$ on validation): a pure cross-window consistency objective is satisfied by a smooth rotation field insensitive to the input. Adding exact relative-rotation supervision from the simulated sequences repaired the rotation (−22.5%) but not the score; simulated supervision *alone* was a null.

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